AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

Good Morning !!!

Silence Hour

Use this time to revise your notes before we begin.

AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

ICT233

Web Technologies and Design.

Lecture Three

Today's Lecture

1.0 Quick Overview of HTML

2.0 HTML Elements and Tags

5.0 Practice: Our First Web Page in HTML

3.0 HTML Document Structure

4.0 Color Coding and HTML Entities

Lecture Objectives

- By the end of the lecture, the student will be able to:
  - Briefly explain HTML.
  - Describe HTML Tags, Elements and Attributes.
  - Identify and explain the structure of HTML5 documents.
  - Specify colors using RGB, HEX, and HSL values.
  - Use entities to represent reserved and special characters in HTML documents.
  - Produce a working HTML5 document and preview it in a web browser of your choice.

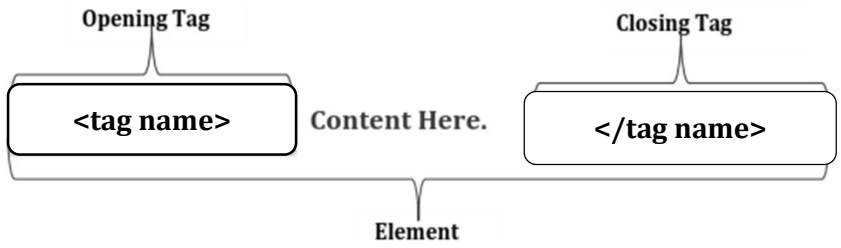
Overview of HTML - 1

- **HTML** stands for **H**yper**T**ext **M**arkup **L**anguage.
- The idea behind HTML was born at the European Laboratory for High Energy Physics (CERN) in Geneva, Switzerland, as early as 1989 .
- A year later, the World Wide Web project was also started there.
- The purpose was to make it easier for scientists at different universities and research laboratories to gain access to research documents of their colleagues .

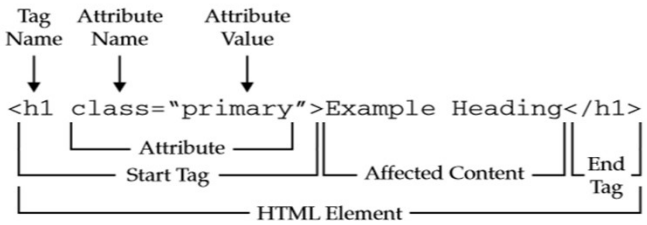
Overview of HTML - 2

- An HTML document is simply a text file(may also contain images, video or audio) that contains the information you want to publish and the appropriate markup instructions indicating how the browser should structure or present the document.
- Markup elements are made up of a start tag, such as ** **, and typically, though not always, an end tag, which is indicated by a forward slash within the tag, such as ** **.
- The tag pair should fully enclose any content to be affected by the element, including text and other HTML markups.

HTML Elements and Tags - 1



A graphical overview of the HTML markup syntax shown so far is presented here:



HTML Elements and Tags - 2

- An HTML **element** consists of both the content and its markup.
- A **tag** consists of the element name(usually an abbreviation of a longer descriptive name) within angle brackets(< >).
- The element name appears in the opening or start tag and again in the closing or end tag preceded by a forward slash(/).

HTML Elements and Tags - 3

- The tags added around contents are referred to as the **markup**.
- It is important to note that an element consists of both the content and its markup(the start and end tags).
- Elements that have content are referred to as **Containers**. e.g ** **Paragraph text** **
- However, not all elements have content, and these are referred to as **Empty elements**.

e.g **<input type="text" name="textbox" />**

<input type="text" name="textbox">

Attributes

- Attributes are instructions that clarify or modify an element.
- Attributes allow us to add extra information to an element.
- The ***img*** element for example requires the ***src*** (short for "source") attribute to provide the location of the image via its path or URL.
- The syntax for adding attributes is as follows:

<element name attribute-name="value">Content</element name>

<element name attribute-name="value" />

<img src="myphoto.jpg" alt="This is my photo" />

<h2 style ="text-align: center; color:#FF0000">

This is a heading content formatted according to the CSS rules specified using the style attribute.

</h2>

HTML Comments

- Just as comments are used in programming to document codes, HTML comments are not displayed by the browser, but they can help document your HTML.
- With comments you can place notifications and reminders in your HTML document.
- The syntax for HTML comments is given below:
- <!-- Your comment goes here -->

e.g <!-- This is a comment in HTML -->

```
<!-- Do not display this at the moment
<img src="pic_mountain.jpg" alt="Mountain">
-->
```


HTML Document Structure

- The way a document is marked up with elements and their attributes is according to a Document Type Definition(DTD).
- DTD are set of rules that govern the way in which a document can be marked up.
- The authoritative source for information about HTML and the HTML DTD is the World Wide Web Consortium(W3C) at http://www.w3c.org.

Basic HTML5 Document Structure

```
<!DOCTYPE html>
    <html lang="en">
    <head>
        <title> Title of web page goes here. </title>
        <meta name= "viewport" content=
          "width=device-width, initial-scale=1.0">
    </head>
    <body>
        Main content of the web page goes here.
        This is what is displayed in the web browser window
    </body>
</html>
```


Detailed HTML5 Document Structure

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <title> The title of the page goes here</title>
        <meta name= "viewport" content= "width=device-width, initial-
          scale=1.0">
        <meta charset="UTF-8">
        <link rel="stylesheet" type="text/css" href="url to CSS file">
        <script type="text/javascript">
                JavaScript  or Jquery codes go here.
        </script>
    </head>
    <body>
       Other HTML5 elements  and tags are used here within the body  tag to
       produce the actual content of the web page.
    </body>
</html>
```

The <!DOCTYPE> Tag

- The Document Type , or doctype for short is indicated by the **<!DOCTYPE>** tag in HTML documents.
- This specifies the rules for the document language so the browser knows how to interpret the HTML code and display it properly.

The <html>…</html> Tag

- The **<html>** **</html>** tags surround the entire HTML document and its referred to as the **root element**.
- The root element is the container element for all other elements in the document
- The opening tag ; **<html>** tells the browser that, whatever content that comes after it should be treated as an HTML content until the browser meets the closing tag **</html>.**

The <head> </head> Tags

- The **<head>** **</head>** tags specify the header segment of any HTML document.
- The **<title>** **</title>** tags are used within the **<head>** **</head>** tags to specify the title of a given web page.
- Other tags such as **<meta>, <link>, <style> and <script>** are also used within the **<head>** tag of an HTML document.

The <body>…</body> Tag

- The <**body**>…**</body>** tag of a web page play an important role with regards to the page's actual content .
- The **body** element contains information about the page's background color, the background image, as well as the text and link colors, and all other items that are displayed on a web page.
- If the **body** element is left blank, web browsers will revert to their default settings.

Creating Syntactically Correct Code

- The following **six(6)** rules are worth following in creating syntactically correct codes:
  1. Documents must be well-formed.
  2. All tags must nest properly and not overlap. e.g ** ** bold text** **
  3. Use all lowercase for element names. e.g: ** **
  4. Always use closing tags. e.g ** **some text** **
  5. Empty elements are marked with a closing forward slash: e.g **
**
  6. Attribute values must be contained in quotation marks: e.g **<img src="myselfie.jpg" />**

Color Codes - 1

- Colors are set using color names, HEX, RGB, HSL, RGBA, and HSLA codes.
- There are several resources available on the Internet that chart colors and their hexadecimal values.
- Check this site for some details:
- https://www.w3schools.com/Html/html_colors.asp

RGB COLOR	HEXADECIMAL VALUE
White	#FFFFFF
Black	#000000
Red	#FF0000
Green	#00FF00
Blue	#0000FF
Magenta	#FF00FF
Cyan	#00FFFF
Yellow	#FFFF00
Aquamarine	#70DB93
Baker's Chocolate	#5C3317
Violet	#9F5F9F
Brass	#B5A642
Copper	#B87333
Pink	#FF6EC7
Orange	#FF7F00

Specifying Color using Hex Values

- In HTML, a color can be specified using a hexadecimal value in the form: #***rrggbb***
- Where **rr (red), gg (green)** and **bb (blue)** are hexadecimal values between 00 and ff (same as decimal 0-255).
- For example, #ff0000 is displayed as red, because red is set to its highest value (ff) and the others are set to the lowest value (00).

Specifying Color using RGB Values

- In HTML, a color can be specified as an **RGB** value, using this formula: **rgb(*red, green, blue*)**
- Each parameter (red, green, and blue) defines the intensity of the color between 0 and 255.
- For example, rgb(0, 0, 255) is displayed as blue, because blue is set to its highest value (255) and the others are set to 0.
- To display the color black, all color parameters must be set to 0, like this: rgb(0, 0, 0).
- To display the color white, all color parameters must be set to 255, like this: rgb(255, 255, 255).

Specifying Color using HSL Values

- In HTML, a color can be specified using hue, saturation, and lightness (HSL) in the form:

  hsl(*hue, saturation, lightness*)
- Hue is a degree on the color wheel from 0 to 360. e.g. 0 is red, 120 is green, and 240 is blue.
- Saturation is a percentage value, 0% means a shade of gray, and 100% is the full color.
- Lightness is also a percentage value, 0% is black, 50% is neither light nor dark, 100% is white.

Specifying Color using names and Codes

```
<style>
   div{
      background-color: #9d00ff;
      color: white;
   }
   span{
      background-color: hsl(277, 100%, 50%);
      color: rgb(255,255,255);
   }

</style>
```


HTML Entities

- HTML entities define a way of representing reserved HTML characters and special characters not present on your keyboard.
- For instance, if you use the less than (<) or greater than (>) signs in your text, the browser might mix them with tags.
- Entities in HTML documents are represented as: &*entity_name*; or &#*entity_number* ;
- E.g **<** or **<** for the less than(<) symbol.

HTML Entities

Result	Description	Entity Name	Entity Number
	non-breaking space		
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	registered trademark	®	®

Source: http://www.w3schools.com/html/html_entities.asp

Creating a Web Page using HTML- 1

1. Choose a text editor(e.g notepad, notepad++, sublime Text).
2. Launch the chosen text editor.
3. In the chosen text editor's working area, type the required elements of a basic HTML document as in the example below.

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <title> My First Web Page</title>
        <meta name= "viewport" content= "width=device-width,
          initial-scale=1.0">
        <meta charset="UTF-8">
    </head>
    <body>
    </body>
</html>
```

Creating a Web Page using HTML- 2

4. Type a text content of your choice within the **<body>** **</body>** tags of the document structure.
5. From the File menu of your chosen text editor, click on Save As and choose a suitable location for saving your file.
6. In the **File name:** text box enter ***"filename.html"*** and in the **Save as type:** combo box, choose **All Files** if you used Notepad or **Hyper Text Markup Language file(*.html;*.htm;*.shtml;*.shtm;*.xhtml;*.hta)** if you used notepad++.
7. Click on the Save button to return to the window of your chosen text editor.
8. Navigate to the location where you saved your ***"filename.html"*** ; it should appear in the icon of the default web browser on your PC with the name ***"filename"*** .
9. Double click on it to preview your page in the browser.

Practice Session - 1

1. Launch a text editor of your choice.
2. Type the required HTML document structure.
3. Type the text below in the body segment of the document structure produced in step 2.

Hi Pals!. HTML is really easy to learn. All you need is a simple text editor and a web browser and you are good to go. The most important thing here is to learn and understand the various tags that are used within the body segment of the entire document structure. Any way, thanks to my Web Technologies and Design lecturer for showing the way!

Practice Session - 2

4. Save the file as firstpage.html a folder you created earlier.
5. Double click on the 'firstpage.html' to preview it in your default browser.
6. Go back to the file 'firstpage.html' and right-click on it and then select edit with…..and choose the text editor you used.
7. Make changes to the file, save it and preview the edited version in your chosen browser.

Focus Questions?

1. Differentiate between HTML tags and HTML elements with examples.
2. Produce a sample HTML page that shows your curriculum vitae using a text editor of your choice and preview the file in a browser of your choice.
3. Find out and read on the evolution of HTML from HTML 1.1 to HTML5.
4. List and explain the new elements in HTML 5.

Our Next Lecture …

1. Formatting HTML documents.
2. Lists in HTML.
3. Creating Links in HTML.
4. Including Multimedia in HTML documents.